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JAIPUR NATIONAL UNIVERSITY

Semester/Year End/Back Examination: December-2023

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B. Tech (Computer Science)
Semester/Year:	I/I
Paper Code:	BTEPHCO103T22
Title of Paper:	Engineering Physics
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. The Question papers consist of two sections.
2. Section A consist of seven questions, attempt any five.
3. Section B consist of six questions, attempt any four.
4. Marks allotted are mentioned against each question.

Question
Number

SECTION-A

Marks
Allotted

- | | | |
|----|--|---------|
| 1. | a) Why fringes circular and center are dark in Newton's ring experiment? Explain? | (2+2=4) |
| | b) How circular polarized light can be changed into plane polarized light? Explain? | |
| 2. | a) Describe Rayleigh's criterion of resolution. | (2+2=4) |
| | b) What is the outcome of double slit experiment? | |
| 3. | a) Define principle of laser action. | (2+2=4) |
| | b) Define temporal and spatial coherence. | |
| 4. | a) Distinguish characteristic between ordinary and laser light. | (2+2=4) |
| | b) A He-Ne laser, giving light at 6330 Å has a coherence length of 20 km. Determine its coherence time and the number of waves per wave train. | |

a) Derive Schrodinger time independent and dependent wave equation. [5+5=10]

b) Deduce relation between phase and group velocity. Find the phase and group velocities of an electron whose de-broglie wavelength is $1.2 \text{ \AA}$.

6. a) Explain single mode and multimode fiber. Give the characteristics of each type of mode. [5+5=10]

b) Write Maxwell's electromagnetic wave equation in free space and also calculate the numerical aperture, acceptance angle and critical angle of the optical fiber if the refractive index of core is 1.5 and refractive index of cladding is 1.45.

describe experimental arrangement
for Newton's rings exp.

describe Lorentz transformation

(a)

5. a) Write down postulates of special theory of relativity. (2)
b) State negative result of Michelson-Morley experiment. (2+2)
6. a) Distinguish between matter wave and electromagnetic wave.
b) Give the reason, why electrons cannot exist inside the nucleus of atom. (2+2)
7. a) Distinguish between Gradient, divergence and curl of vector.
b) What is total internal reflection of light?

SECTION-B

1. a) Derive an expression for the intensity distribution of fraunhofer diffraction due to single slit. [5+5=10]
b) What is double refraction phenomenon? Newton's rings are observed by keeping a spherical surface of 100 cm radius on a plane glass plate. if the diameter of the 15th bright ring is 0.590 cm and the diameter of the 5th ring is 0.336cm. What is the wavelength of the light used. ←
2. a) What are Einstein coefficients A and B? Establish a relation between them. [5+5=10]
b) Describe the construction and working of He-Ne laser.
3. a) Deduce Einstein's mass-energy relation. [5+5=10]
b) What is time dilation. Calculate the percentage contraction of a rod moving with velocity of 0.8c in the direction inclined at 60° to its length?
4. a) Show that $x^2 + y^2 + z^2 - c^2 t^2$ is invariant under Lorentz transformation. [5+5=10]
b) A rocket is chasing enemy's space ship. An observer on the earth observes the speed of rocket to be 2.5×10^8 m/s and that of space ship 2×10^8 m/s. Calculate (i) the velocity of enemy's ship as seen by rocket. (ii) The velocity of rocket as seen by enemy's ship. ←